

Problem 1. $M^* = \{ID_0, ID_1, \dots, ID_{149}\} \cup \{\text{Alarm}, \text{Lockdown}\}$ SYDNEE BOOTHBY
 $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$, perfectly secure over $\mathbb{Z}_n$ CPTS 327 02/17
 $\text{Enc}(K, m) = (m + K + 7) \% n \quad m \in M^* \quad f: M^* \rightarrow \mathbb{Z}_n$

a. n must be at least 152 (each value in M^* can be uniquely represented)
 $\hookrightarrow c \quad n \geq 152$ (for perfect security)

b. $F(m)$: $F(ID_i) = i$ for each $i \in \{0, \dots, 149\}$
 emergency symbols, map $\{\text{Alarm}, \text{Lockdown}\}$ to set:

$\hookrightarrow b. \{150, 151\}$ to ensure mapping is injective (each message in M^* is unique)

// consider instantiation $n=500$ for underlying scheme, while mapping function f restricted to 152 indices.

c. Key space $\mathbb{Z}_{500}$ meets requirements for perfect security:

$\hookrightarrow A$, Yes, each message is still mapped one-to-one (all unique ciphertexts)

d. For fixed $m \in M^*$, as long as K is sampled uniformly from $\mathbb{Z}_{500}$, c is

$\hookrightarrow D$: mod on $n=500$ allows for any value from 0 to 499, as the key is uniformly distributed from 0 to 500, mod 500 allows for all values within that range

2. $\Sigma = (\text{Gen}, \text{Enc}, \text{Dec})$ over message space $M = \mathbb{Z}_6$ and $C = \mathbb{Z}_6$
 $\text{Enc}(K, m) = (m + K) \% 6$

a. $\text{Adv}(A_{\text{alice}}) = [\underbrace{\Pr[\text{Acc}]=1 \mid b=1]}_{\text{prob. output odd and in world 1}}] - \underbrace{\Pr[\text{Acc}]=1 \mid b=0]}_{\text{prob. output odd and world 0}}$

$$= \underbrace{m_1=1, K=0,2,4}_{0.45+0.10+0.05=0.60} \quad \underbrace{m_1=0, K=1,3,5}_{0.25+0.10+0.05=0.40}$$

$$\text{Adv}(A) = 0.60 - 0.40 = 0.20$$

b. $\text{Adv}(B) = \underbrace{\Pr[\text{Acc}=1 \mid b=1]}_{\text{world 1}} - \underbrace{\Pr[\text{Acc}=1 \mid b=0]}_{\text{world 0}}$

$$\begin{array}{ll} \text{output 1 if } m+K \% 6 = 1 \text{ or } 2, \text{ and } m=1 & \text{output 1 if } m+K \% 6 = 1 \text{ or } 2, \text{ and } m=0 \\ \Pr\left\{ \begin{array}{l} 1=1+K \\ 2=1+K \end{array} \right\} K=0,1 & \Pr\left\{ \begin{array}{l} 1=0+K \\ 2=0+K \end{array} \right\} K=1,2 \\ 0.45+0.25=0.70 & 0.25+0.10=0.35 \end{array}$$

$$\text{Adv}(B) = 0.70 - 0.35 = 0.35$$